# INTRODUCTION

## Why are biological and ecological traits of weeds important?

It is easy to focus on understanding and implementing weed control tactics. However, it is also important to consider aspects of weed biology and ecology. Ongoing weed control cannot be successful until we understand how weeds grow and interact with other species, including crops, management practices and the environment.

Through having increased knowledge of the biology and ecology of weeds, growers will have the tools to implement the most appropriate management strategies for the weed spectrum occurring on their properties. By being aware of weed ecology, the likely weed-related implications of changing farming practices such as new crops, earlier sowing, or soil amelioration can be understood.

Weed scientists and advisers can offer advice on weed control, but it is important to remember individual weed species can behave quite differently across the breadth of Australia. The only way to accurately judge what an individual weed population is doing is to look at it, fairly regularly, and to understand why the weeds are acting or reacting in certain ways.

For each weed in this book, the authors provide information under seven headings, which are discussed below. This introduction explains why each of these sections is important and how each relates to weed control.

## Background

The first section considers the name of the weed species and a map of where it is found. Weed species can have many different common names and sometimes may also have different scientific names. This can make identification difficult, but weed control can be compromised by not knowing what weed it is. However, for some species, identification is more important than others. For example, it is very difficult to differentiate between the barley grass species *Hordeum leporinum* or *H. glaucum* without specialist taxonomic knowledge and a magnifying glass. However, many herbicide labels just refer to 'barley grass' without specifying a species, because the product will work equally well on both species.

## Description

A description of the weed helps with accurate identification, and this section also highlights similar species that the weed may be confused with. However, the description of the weed also helps to decide which management practices are needed. Is the species a winter or summer annual? This determines when control tactics need to occur. Is the growth of the plant flat (prostrate) or erect? This will influence how competitive a species is with the crop and if harvest weed seed control is possible.

### Why is it a weed?

Not every weed is equally bad in every scenario. Capeweed, for example, is one of Western Australia's most common species, but this relatively small, prostrate species is rarely a major economic concern for grain growers. A toxic species is probably a greater problem in an enterprise with livestock than cropping alone. Conversely, a non-toxic, palatable species is likely to be less of an issue in a livestock-dominated enterprise.

## Seed

### Dormancy and germination

The best time to control weeds is when they are seeds or seedlings; at this stage they are easier to kill and have not caused economic damage. However, it is necessary to understand seed dormancy and germination requirements to know when seedlings will emerge. Initial seed dormancy (after-ripening) determines when a seed will germinate and emerge in the year after seed-set. For example, how many of the seeds of a winter annual weed stay dormant over summer and how many emerge in response to summer rainfall? Germination biology also influences seed germination and seedling emergence. Do the seeds emerge immediately after rainfall events, like sowthistle? Do the seeds need cold temperatures to trigger emergence? Are the seeds sensitive to light or burial, leading to increased emergence following minimum tillage (with moderate seed burial) compared with no-tillage systems (with less than five per cent seed burial)? Aspects of seed biology need to be considered in terms of the broader agronomic system.

### Seedbank persistence

How long does the seedbank last? This is a key question in planning an integrated weed management program. Weeds that are more likely to germinate and degrade in one to two years will require different management than weeds more likely to persist in the seedbank for three years or more. However, this can be influenced by the environment. Has the seed been buried? Is the site high or low rainfall? Will the seeds attract granivorous (seed eating) species? Many different factors will influence the dormant seedbank and determine how many years of management are required to get the seedbank to low levels.

### Seed production and dispersal

Most weeds have the potential for very high seed production but there can still be more than a 10-fold difference in the maximum seed-set between weed species. However, the timing of seed production and the timing of seed shed and seed dispersal mechanisms all influence management strategies. Do in-crop weeds retain sufficient seed to make harvest weed seed control worthwhile? Do summer weeds set seed quickly and need immediate control, or are there a few weeks of vegetative growth before control actions are needed to ensure complete prevention of seed-set? Is seed dispersal so common and widespread that management programs need to consider what is happening in neighbouring areas, or is it acceptable to just focus on a single paddock?

## Competition

How well a weed species competes with the crop is the primary question for most Australian growers. Tall, vigorous annual species such as brome grass and feathertop Rhodes grass are consistently detrimental to yield. Prostrate species, or species with initial emergence later than the crop, such as doublegee, often have less of an impact on crop yield. Summer weeds, in a region with only winter annual crops, in seasons with small rainfall events on sandy soil (poor water-holding capacity) may have little impact on subsequent crop yield.

Table 1 contains a summary of the characteristics of the weed species included in this publication.